Hyperspectral Image Classification With Contrastive Graph Convolutional Network

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***Abstract*— Recently, graph convolutional network (GCN) has been widely used in hyperspectral image (HSI) classification due to its satisfactory performance. However, the number of labeled pixels is very limited in HSI, and thus, the available supervision information is usually insufficient, which will inevitably degrade the representation ability of most existing GCN-based methods. To enhance the feature representation ability, in this article, a GCN model with contrastive learning is proposed to explore the supervision signals contained in both spectral information and spatial relations, which is termed contrastive GCN (ConGCN), for HSI classification. First, in order to mine sufficient supervision signals from spectral information, a semisupervised contrastive loss function is utilized to maximize the agreement between different views of the same node or the nodes from the same land cover category. Second, to extract the precious yet implicit spatial relations in HSI, a graph generative loss function is leveraged to explore supplementary supervision signals contained in the graph topology. In addition, an adaptive graph augmentation technique is designed to flexibly incorporate the spectral–spatial priors of HSI, which helps facilitate the subsequent contrastive representation learning. The extensive experimental results on six typical benchmark datasets firmly demonstrate the effectiveness of the proposed ConGCN in both qualitative and quantitative aspects.**

***Index Terms*— Contrastive learning, graph augmentation, graph convolutional network (GCN), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL image (HSI) classification plays an increasingly significant role in environmental monitoring, precision agriculture, mineral identification, and land cover classification. Different from conventional remote sensing images, HSI is composed of diverse contiguous spectral bands, providing detailed spectral information in addition to spatial relations. This property is beneficial to attributing each pixel of HSI to a certain category.

In the past few decades, various methods have been proposed for HSI classification. The early staged methods are mainly based on conventional pattern recognition methods. In spite of the achievements obtained by these methods, they suffer from handcrafted configurations. To avoid intricate feature engineering, deep learning-based methods can obtain high-level representations via gradually aggregating the low-level features and have attracted increasing attention in recent years. As one of the most representative methods, convolutional neural networks (CNNs) have achieved state-of-the-art performance due to their powerful generalization ability. However, the receptive field of CNN is restricted by a regular square area, and thus, the CNN fails to adaptively capture the geometric variations of different land cover categories in HSI. To deal with this deficiency, recently, a number of graph convolutional network (GCN)-based HSI classification methods [1], [2], [3], [4], [5] have been proposed and achieved promising performance. For example, in [1], a multiscale dynamic GCN (MDGCN) was presented to obtain the dynamic graph and explore spectral–spatial correlations at multiple scales. Based on the MDGCN, to tap into the advantage of contextual information, a context-aware dynamic GCN [2] was proposed to further mine node features for HSI classification.

However, the representation ability of most existing GCN-based methods is still limited due to the following defects. First, the number of labeled pixels in HSI is usually insufficient in practical scenarios [6], [7], [8], [9]. The supervision information provided by few labeled pixels is insufficient to train a GCN model with good discriminative ability [3]. Second, current GCN-based methods fail to exploit the long-range dependencies among image regions, as they mainly aim at encoding the pairwise importance among local image regions, which could lead to suboptimal performance [10].

To accommodate the abovementioned issues, we propose a GCN-based method for HSI classification, termed "contrastive GCN" (ConGCN). First, we aim to sufficiently extract the supervision signals carried by the hyperspectral data themselves for network training. Concretely, we propose to use contrastive learning [11], [12] to encode the similarities among the spectral signatures of image regions. Contrastive learning is one of the representative self-supervised learning approaches and is able to generate data representations by

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learning to encode the similarities or dissimilarities among unlabeled examples. It has emerged as a powerful technique for graph representation learning recently [13], [14]. Different from the contrastive learning-based methods used in vision representation learning [11], [12], we design a semisupervised contrastive loss to jointly extract supervision information from the scarce yet valuable labeled data and the rich unlabeled data. To flexibly obtain different graph views for the contrastive learning model, we devise an adaptive graph augmentation technique by incorporating the spectral–spatial priors of HSI. In addition to the supervision signals contained in spectral features, the spatial relations among image regions are nonnegligible in HSI. Consequently, we develop a graph generative loss to model the relationship among spatially neighboring regions. This can be considered as a supplement to the supervision information extracted from spectral signatures. As such, the originally scarce supervision signals can be further enriched by exploring the knowledge from both spectral information and spatial relations of HSI, thereby leading to enhanced data representations. Besides, considering that conventional GCN models can only capture the localized dependencies among image regions [2], we propose to employ the localized and hierarchical graph convolution simultaneously. By doing so, both the local and global contextual information of HSI can be incorporated to help produce comprehensive representations.

In conclusion, the proposed ConGCN can exhaustively explore the supervision information contained in both spectral signatures and spatial relations of HSI by introducing the semisupervised contrastive learning and graph generative loss. As a result, the abundant supervision information from hyperspectral data can facilitate the contrastive model training. Here, the spectral–spatial priors in HSI are exploited for adaptive data augmentation to produce meaningful contrastive pairs. Moreover, the contextual information at different spatial levels is jointly incorporated by employing localized and hierarchical graph convolution, which further helps enhance the representation power of the contrastive learning model.

In summary, the contributions of our work are as follows.

1) To the best of our knowledge, this is the first work to adopt a ConGCN model for HSI classification. Different from the existing GCN-based methods, which require sufficient label information for network training, our proposed ConGCN can exhaustively exploit the supervision information from hyperspectral data themselves to train the networks. Hence, the proposed method reveals a new type of GCN model that can perform well with very limited labeled pixels.
2) The semisupervised contrastive loss and graph generative loss are designed to exploit the supervision signals contained in the spectral domain and spatial relations of HSI, respectively.
3) By performing localized and hierarchical graph convolution simultaneously, both local and global contextual information of HSI can be leveraged for expressive representation learning.
4) We devise an adaptive graph augmentation technique by incorporating spectral–spatial priors, which helps boost the performance of contrastive learning.

The structure of this article is organized as follows. Section II reviews related typical prior works. Section III describes the pipeline of our proposed ConGCN method. Then, the proposed method is presented in Sections IV–VI, which focuses on adaptive graph augmentation, localized and hierarchical graph convolution, and the proposed loss functions, respectively. In addition, Section VII shows the experimental results of different methods on four widely used datasets. Finally, Section VIII concludes this article.

II. Related Works

This section reviews some typical prior works related to this article, which include HSI classification methods, GCN models, and contrastive learning approaches.

A. HSI Classification Methods

During the past few decades, abundant methods have been put forward for HSI classification. The early staged methods were mainly anchored on conventional machine learning methods, such as kernel-based methods [15], [16], recursive filtering [17], k-nearest-neighbor classifier [18], support vector machine (SVM) [19], and Markov random field [20]. Unfortunately, they usually relied on empirically designed handcrafted features, so their performances are often far from perfect.

To tackle this problem, a number of deep methods, such as generative adversarial networks (GAN) [21], [22] and CNN [23], [24], [25], [26], have been further employed to promote HSI classification. For example, Feng et al. [21] introduced self-supervised learning to GAN for HSI classification by exploring abundant unlabeled examples, which helped improve the generation and classification abilities of the generator and discriminator, respectively. To reduce the dimensionality of HSI during classification, Feng et al. [26] developed an end-to-end CNN combining band selection, feature extraction, and classification for HSI. Despite the fact that the CNN-based methods display encouraging performances in some cases, they still have several defects. First of all, the receptive field of CNN is a regular square area; therefore, CNN-based methods are unable to adaptively capture or perceive the geometric variations of different land cover categories in HSI. Besides, the weights of each convolution kernel are identical in different spectral bands. Consequently, the details of land cover boundaries are probably lost after feature abstraction, so the pixels around boundaries are likely to be misclassified due to the inflexible convolution kernel. As a sequel, more and more GCN-based HSI classification methods are proposed to address the above problems, which will be introduced below.

B. GCN and Its Application to HSI Classification

Graph neural network (GNN) [27] mapped the graph or its vertices to an Euclidean space via a transfer function. Owning to this, GNN is capable of processing graph-structured non-Euclidean data, therefore showing better flexibility and adaptability than CNN. After Bruna et al. [28] introduced graph Laplacian matrix for graph convolution, many GCN models have been proposed and obtained promising performance.

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Among these methods, Kipf and Welling [29] innovatively applied GCN to semisupervised learning, which approximated spectral graph convolutions in a localized region, so that the proposed method was able to learn hidden representation via encoding both graph structure and node features. However, it belongs to transductive methods, which fails to classify unseen or newly added nodes. To tackle with this issue, Hamilton et al. [30] developed an inductive framework termed graph sample and aggregate (GraphSAGE) via generalizing the simple graph convolution to trainable aggregation functions.

For HSI classification, GCN is able to capture and preserve the boundaries of different land cover categories flexibly due to its good ability in processing non-Euclidean data. As a result, GCN-based HSI classification methods have emerged and attracted a lot of attention. To the best of our knowledge, Qin et al. [31] were the first to introduce GCN into HSI classification, which leveraged the features of both adjacency nodes in graph and the neighbor pixels in the HSI. To avoid the imprecise initial graph, Wan et al. [1] proposed to refine the graph gradually during the convolution process, so graph convolution was operated on a dynamic graph rather than a predefined fixed graph. To sufficiently explore the contextual information, Wan et al. [2] proposed the context-aware dynamic GCN to capture relations among the regions originally far away in the original spatial positions. To overcome the drawback of the manually constructed graph, Wan et al. [3] developed a new dual interactive GCN to adaptively learn a discriminative region-induced graph. In [4], a semisupervised nonlocal GCN was developed to exploit labeled and unlabeled data, simultaneously. Moreover, Bai et al. [5] developed a deep attention GCN to mine HSI features via focusing on the spectral information, which has a large impact on classification. Liu et al. [32] leveraged CNN to generate pixel-level features to enrich the superpixel-level features of GCN, thus simultaneously taking the advantages of CNN and GCN. To fuse different-scale features from coarse to fine, Liu et al. [33] merged adjacent superpixels and then converted them into multilevel graphs. Then, a multilevel graph network was employed to exploit features of HSI from a hierarchical perspective.

However, the generalizability of most existing methods is usually limited due to their transductive setting. To solve this problem, Ding et al. [34] proposed a multiscale graph network combining GraphSAGE with context-aware learning to understand the global and local information in a graph. However, it only utilized spectral features to construct the graph. To incorporate the relationship between adjacent nodes at the stage of graph construction, Guo et al. [35] put forward a dual graph U-net via integrating spatial graph and spectral graph simultaneously.

Nonetheless, the representation ability of most existing GCN-based methods is still limited due to the inadequate labeled examples. To enhance the representation ability, in our work, we explore the supervision signals contained in spectral information by employing contrastive learning, as well as spatial relations by deploying GCN. Hence, the proposed method can obtain better classification results than existing methods.

C. Contrastive Learning

Contrastive learning is one of the self-supervised representation learning methods, which aims to obtain discriminative feature representation by leveraging the similarities and dissimilarities between examples [36]. It has gained increasing attention due to the promising results in various fields, such as computer vision [11], [12] and natural language processing [37].

Simple framework for contrastive learning (SimCLR) [11] is a well-known contrastive learning method and is anchored on a Siamese network that learns latent representations via maximizing agreement between differently augmented views of the same example. Similar to SimCLR, momentum contrast (MoCo) [12] utilized information maximizing noise contrastive estimation (InfoNCE) loss [38] and stored negative examples by using a memory bank rather than a large mini-batch size.

The marriage of contrastive learning and GCN has also been observed in recent years. For example, Velickovic et al. [39] proposed to learn graph representations in an unsupervised manner via classifying local–global pairs and negative-sampled counterparts. Besides, to address the challenge of data heterogeneity in graphs, You et al. [40] developed graph contrastive learning with four types of graph data augmentations, each of which imposed certain prior on graph data and encoded the extent and pattern. Apart from this work, Peng et al. [41] introduced a novel concept of graphical mutual information for graph representation learning, which generalized the idea of conventional mutual information computations from vector space to the graph domain.

Nevertheless, these methods are not applicable to HSI classification studied in this article, as they fail to consider the spatial and spectral cues, which are naturally and uniquely inherited by HSIs. Besides, they also ignore the class information carried by a handful of labeled data for HSI classification. Consequently, in this article, we devise a new ConGCN, which is able to acquire discriminative representations for accurate HSI classification.

To the best of our knowledge, there have been two works [42], [43] employing contrastive learning for HSI classification. However, they simply use the traditional paradigm of contrastive learning by utilizing unlabeled examples for pretraining and fine-tuning the model with few labeled examples. In order to improve the discriminative power of the generated representations, our proposed method additionally incorporates the available class information by using a semisupervised contrastive loss function. Furthermore, our proposed method incorporates spatial relations to the contrastive objective. As a result, the proposed ConGCN can produce more effective feature representations than ResNet-50 in [42] and the transformer model in [43]. The advantage of our ConGCN has also been empirically demonstrated in Section VII.

III. Pipeline of the Proposed Method

This section describes the pipeline of our proposed ConGCN method (see Fig. 1). When an input HSI is given, it is first segmented into a set of compact image regions

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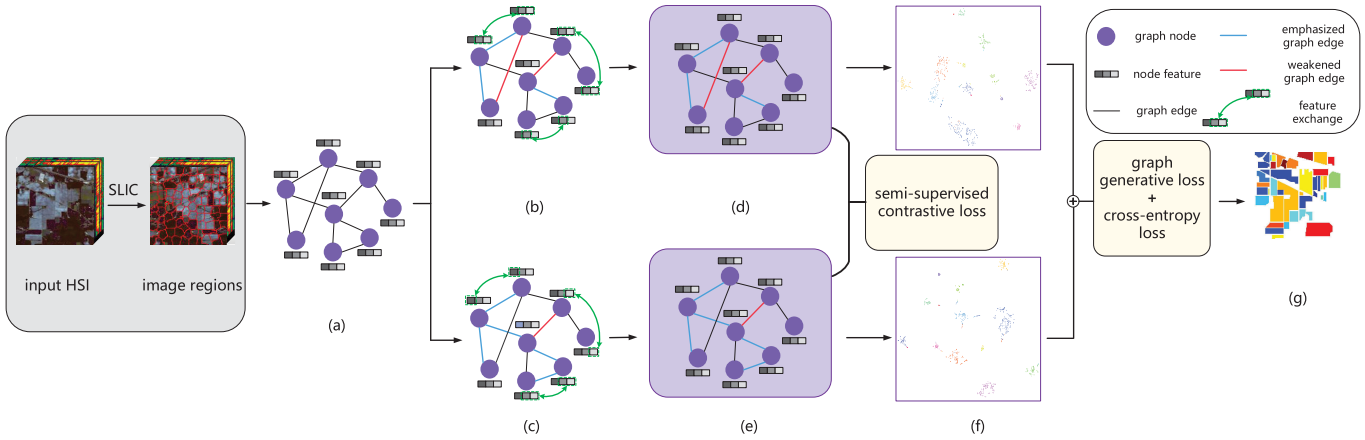

Fig. 1. Framework of our proposed ConGCN. In this figure, purple circles and black lines represent graph nodes and edges, respectively. (a) Graph $\mathcal{G}$ is constructed over the image regions produced by the SLIC algorithm. (b) and (c) Notations $\tilde{\mathcal{G}}_1$ and $\tilde{\mathcal{G}}_2$ denote two different graph views generated by the proposed adaptive graph augmentation technique. The blue lines and red lines represent emphasized edges and weakened edges, respectively. Besides, the green arrows denote the exchange of node features. (d) and (e) Localized graph convolution and hierarchical graph convolution, respectively. The semisupervised contrastive loss is computed across the two graph views. (f) Node representations learned from the two views. (g) Classification result is acquired via integrating the learned representations. Here, cross-entropy loss is used to penalize the difference between the output and the labels of the initially labeled seed superpixels. In the meanwhile, the graph generative loss is employed to exploit the topological information of the graph.

by using the Simple Linear Iterative Clustering (SLIC) algorithm [44]. Next, a graph (i.e., $\mathcal{G}$) is constructed by treating each of the image regions as a graph node. After that, the proposed adaptive graph augmentation (Section IV) is conducted on this graph to obtain two augmented graphs (i.e., $\tilde{\mathcal{G}}_1$ and $\tilde{\mathcal{G}}_2$). Finally, localized and hierarchical graph convolutions (Section V) are performed based on the two augmented graphs to obtain node representations, where the semisupervised contrastive loss (Section VI-A) and the graph generative loss (Section VI-B) are utilized for network training (Section VI-C).

As mentioned above, the classical SLIC [44] algorithm is adopted to segment the entire HSI into a number of compact superpixels. This is because there are numerous image pixels in an HSI, and thus, constructing a pixel-level graph could be extremely time-consuming. By using SLIC, each superpixel clustering a set of homogeneous pixels with strong spectral–spatial similarity is treated as a graph node. Here, the node features correspond to the average spectral signatures of the pixels involved in the corresponding superpixel. Hence, the number of graph nodes can be significantly reduced to improve the computational efficiency. Another benefit of using the superpixel segmentation is that the generated image regions can preserve the local structural information of HSI, as adjacent pixels with high spatial consistency usually have a large probability to belong to the same land cover category.

In our proposed GCN-based method, an undirected graph made up of $n = l + u$ nodes is defined as $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$, where $\mathcal{V} = \{\mathbf{x}_1, \ldots, \mathbf{x}_l, \mathbf{x}_{l+1}, \ldots, \mathbf{x}_n\}$ is the node set containing all examples (i.e., superpixels), and $\mathcal{E}$ is the edge set modeling the connectivity among the nodes. In $\mathcal{V}$, the first l nodes are labeled, while the remaining u nodes are unlabeled. Note that the label of each superpixel is determined by the most frequent label within this superpixel. The adjacency matrix of $\mathcal{G}$ denoted as $\mathbf{A}$ can be calculated as follows:

$$\mathbf{A}_{ij} = \begin{cases} e^{-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|_2^2}, & \text{if } \mathbf{x}_i \in \mathcal{N}(\mathbf{x}_j) \text{ or } \mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases}, \quad (1)$$

where γ is a temperature parameter and is set to 0.2 according to [1], [2], and [3], $||\mathbf{x}_i - \mathbf{x}_j||_2$ calculates the Euclidean distance between the graph nodes $\mathbf{x}_i$ and $\mathbf{x}_j$, and $\mathcal{N}(\mathbf{x}_i)$ is the neighbor set of $\mathbf{x}_i$. Other important symbols used throughout this article are listed in Table I.

TABLE I
SYMBOLS USED THROUGHOUT THIS ARTICLE

Symbols	Description
$\mathbf{X}$	feature matrix of graph $\mathcal{G}$
$\mathbf{X}_{\text{labeled}}$	feature matrix of labeled examples
$\mathbf{Y}$	label matrix of graph $\mathcal{G}$
$\mathbf{Z}$	network output
$\mathbf{x}_i$	the i-th node in graph $\mathcal{G}$
d	feature dimension of $\mathbf{x}_i$
D	generalized Mahalanobis distance
I	mutual information

IV. Adaptive Graph Augmentation

The core of contrastive learning is to maximize the agreement between differently augmented views of the same example, where data augmentation turns out to be an important prerequisite in contrastive learning. Nevertheless, how to obtain an augmentation technique that is beneficial to graph representation learning remains a challenge due to the non-Euclidean properties of graph-structured data. To address this issue, we propose an adaptive graph augmentation technique, which is able to incorporate spectral–spatial priors in HSI to boost the performance of contrastive learning. Inspired by Zhu et al. [45], the principle of our proposed graph augmentation method is to preserve important structures

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and attributes while perturbing possibly unimportant edges and features. The reason is that it will guide the model to ignore the noise introduced by unimportant edges and features, thus helping to learn important patterns underneath the input graph. To evaluate the importance of edges and features, we leverage generalized Mahalanobis distance [2] and mutual information [46], respectively, which will be later explained.

The proposed augmentation technique for each graph view is made up of the spatial-level graph augmentation (Section IV-A) and the spectral-level graph augmentation (Section IV-B). In the spatial-level graph augmentation, we adaptively emphasize the important edges while weakening unimportant edges, to preserve the intrinsic structure of the graph topology. In the spectral-level graph augmentation, we adaptively exchange the features of adjacent nodes, to explore diverse contexts across different graph views for contrastive learning and help the network obtain the improved representations.

A. Spatial-Level Graph Augmentation

In our proposed method, graph edges reflect the spatial relations among image regions and also constitute the topological structure of data. As a result, the edges are critical in ConGCN to yield good representations. However, most existing graph contrastive methods neglect the intrinsic topological properties of graphs when performing graph augmentation, since they uniformly drop the graph edges, which could result in suboptimal performance. For example, removing some influential edges will deteriorate the graph topology and, thus, result in inaccurate representations. To tackle this problem, we aim to adaptively emphasize or weaken edges according to their influences in the graph.

First, we construct a generalized Mahalanobis distance D to measure the distance between each pair of nodes, which is formulated as follows:

$$D(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{W}_\mathrm{D} \mathbf{W}_\mathrm{D}^\top (\mathbf{x}_i - \mathbf{x}_j)}, \tag{2}$$

where $\mathbf{W}_\mathrm{D}$ denotes a trainable weight matrix. This distance could be utilized to estimate the edge influence between each pair of nodes. Specifically, a small distance $D(\mathbf{x}_i, \mathbf{x}_j)$ often corresponds to a minor edge influence of the edge between $\mathbf{x}_i$ and $\mathbf{x}_j$. Anchored on (2), to adaptively emphasize or weaken the edges, we modify the original adjacency matrix $\mathbf{A}$ by adding an auxiliary matrix $\mathbf{A}'$, which can be calculated as follows:

$$\mathbf{A}'_{ij} = \begin{cases} -\min(e^{D(\mathbf{x}_i,\mathbf{x}_j)-\tau}, e^\tau) + 1, & \text{if } D(\mathbf{x}_i, \mathbf{x}_j) > \tau \\ e^{-D(\mathbf{x}_i,\mathbf{x}_j)+\tau} - 1, & \text{if } D(\mathbf{x}_i, \mathbf{x}_j) \le \tau \end{cases}, \tag{3}$$

where $\mathbf{x}_i \in \mathcal{N}(\mathbf{x}_j)$ or $\mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i)$, $\tau > 0$ is a learnable parameter, and the truncation parameter e^τ is used to avoid excessively large values. Visual explanation of (3) is shown in Fig. 2, where τ is a threshold. As can be observed in Fig. 2, when $D(\mathbf{x}_i, \mathbf{x}_j) \le \tau$, the edge between $\mathbf{x}_i$ and $\mathbf{x}_j$ may have a large influence on the graph and, thus, could be emphasized. In contrast, when $D(\mathbf{x}_i, \mathbf{x}_j) > \tau$, the corresponding edge could be weakened, considering its minor influence on the graph. In addition, the function curve is truncated to $-e^\tau + 1$ when $D(\mathbf{x}_i, \mathbf{x}_j) > 2\tau$, to avoid excessively small values. To construct a randomly augmented graph view, we then generate a random masking matrix $\tilde{\mathbf{R}} \in \{0, 1\}^{n \times n}$. Here, the element $\tilde{\mathbf{R}}_{ij} = \tilde{\mathbf{R}}_{ji}$ is sampled from a Bernoulli distribution $B(1, p_\mathrm{sample})$ if $i \neq j$, and the value of each diagonal element $\tilde{\mathbf{R}}_{ii}$ is fixed to one. The hyperparameter p_sample denotes the probability of performing edge emphasizing or weakening. Afterward, the adjacency matrix $\tilde{\mathbf{A}}$ of the augmented graph can be computed as follows:

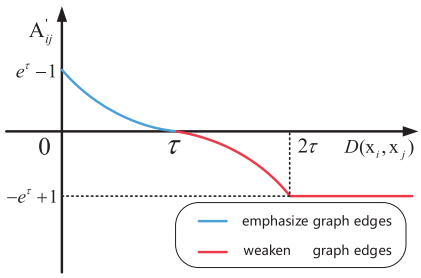

Fig. 2. Visual explanation of (3). The horizontal axis corresponds to $D(\mathbf{x}_i, \mathbf{x}_j)$, and the vertical axis corresponds to $\mathbf{A}'_{ij}$. When $\mathbf{A}'_{ij} > 0$ (drawn in blue), the original graph edges will be emphasized as the value of $\mathbf{A}'_{ij}$ is positive; when $\mathbf{A}'_{ij} < 0$ (drawn in red), the original graph edges will be weakened as the value of $\mathbf{A}'_{ij}$ is negative.

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{A}' \circ \tilde{\mathbf{R}}, \tag{4}$$

where $\circ$ denotes the Hadamard product. By using (4), the spatial-level graph augmentation can be performed on the randomly sampled edges. Next, for $\mathbf{x}_i$ and $\mathbf{x}_j$ satisfying $\mathbf{x}_i \in \mathcal{N}(\mathbf{x}_j)$ or $\mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i)$, the corresponding element $\tilde{\mathbf{A}}_{ij}$ can be normalized to [0, 1] by

$$\tilde{\mathbf{A}}_{ij} \leftarrow \frac{\tilde{\mathbf{A}}_{ij} - \tilde{\mathbf{A}}_\mathrm{min}}{\tilde{\mathbf{A}}_\mathrm{max} - \tilde{\mathbf{A}}_\mathrm{min}}, \tag{5}$$

where $\tilde{\mathbf{A}}_\mathrm{min}$ and $\tilde{\mathbf{A}}_\mathrm{max}$ are the minimum and the maximum values of $\tilde{\mathbf{A}}_{ij}$, respectively. With (5), we can make all the values of $\tilde{\mathbf{A}}$ no less than 0. Therefore, by employing the spatial-level augmentation, the graph edges can be adaptively emphasized or weakened according to their corresponding importance.

B. Spectral-Level Graph Augmentation

Different from the spatial-level graph augmentation that focuses on the spatial relations of image regions in HSI, our proposed spectral-level augmentation aims to perturb the spectral signatures (i.e., node features) for graph augmentation. To achieve this target, [39] uniformly shuffles node features for graph augmentation, which actually treats all feature dimensions equally and ignores their distinct contributions. Inspired by chromosomal crossover [47], [48], [49] in biology, we come up with a new augmentation method that exchanges partial dimensions of node features according to the spectral information of HSI.

Chromosomal crossover represents the exchange of genetic material at the stage of sexual reproduction between two homologous chromosomes' nonsister chromatids, which leads

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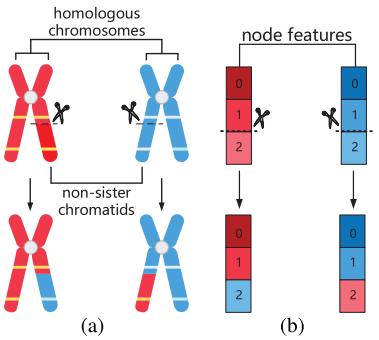

Fig. 3. Analogy of chromosomal crossover and feature exchange. (a) Illustration of the exchange of genetic material during sexual reproduction between two homologous chromosomes' nonsister chromatids. (b) Illustration of spectral-level graph augmentation. In (b), the colored small rectangles represent node features, where "0," "1," and "2" in the figure represent the IDs of different dimensions. Here, the features are exchanged at the dimension with ID "2."

to recombinant chromosomes and significantly increases genetic diversity [50]. Schematic of chromosomal crossover is shown in Fig. 3(a). For spectral-level graph augmentation, we treat two adjacent nodes as homologous chromosomes' nonsister chromatids and regard their node features as genetic material. Then, partial exchange of node features is performed across the graph edges, which is shown in Fig. 3(b). The aim of feature exchange is to provide diverse perturbations for contrastive learning and guide the network to obtain improved representation. Notably, the feature exchange is limited in the adjacent nodes (i.e., the nodes connected by the edges of $\mathcal{G}$) to avoid corrupting the essential structure of graphs.

Motivated by Zhu et al. [45], features in influential dimensions are expected to be exchanged with small probability, as they often carry valuable spectral information that is beneficial to accurate classification. In the proposed spectral augmentation, we first generate a random vector $\mathbf{m}$ with the hth element $\mathbf{m}_h$ $(h = 1, 2, \ldots, d)$ drawn from a Bernoulli distribution independently, namely, $\mathbf{m}_h \sim B(1, 1 - p_h)$, where the parameter p_h reflects the importance of the hth feature dimension. Afterward, feature exchange between each pair of nodes will be performed at the hth dimension if $\mathbf{m}_h = 1$. Here, the parameter p_h is obtained by mutual information, since the mutual information $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ is able to characterize the relationship between the hth feature dimension of $\mathbf{X}_{\text{labeled}}$ (i.e., $\mathbf{X}_{\text{labeled}[:,h]}$) and the label matrix $\mathbf{Y}$. The notation $\mathbf{X}_{\text{labeled}}$ denotes the features of l labeled examples. The notation $\mathbf{Y} \in \mathbb{R}^{l\times c}$ represents the label matrix, where c is the number of land cover categories. The element $\mathbf{Y}_{ij} = 1$ if the ith labeled example belongs to the jth land cover category, and $\mathbf{Y}_{ij} = 0$, otherwise. However, it could be difficult to directly calculate $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$, as the probability distribution of $\mathbf{X}_{\text{labeled}[:,h]}$ is unavailable. Fortunately, according to [46], $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ can be estimated by partitioning them into bins of finite size and approximated via the finite sum. As a result, we partition $\mathbf{X}_{\text{labeled}[:,h]}$ and $\mathbf{Y}$ into N bins, where $n_x(i)$ and $n_y(j)$ represent the number of examples falling into the ith bin of $\mathbf{X}_{\text{labeled}[:,h]}$ and the jth bin of $\mathbf{Y}$, respectively. The notation $n(i, j)$ denotes the number of examples in their intersection. Consequently, $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ can be estimated as follows:

$$I\left(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y}\right) \approx \sum_{i,j} p(i, j) \log \frac{p(i, j)}{p_x(i)p_y(j)}, \tag{6}$$

where $p_x(i) \approx (n_x(i)/N)$, $p_y(j) \approx (n_y(j)/N)$, and $p(i, j) \approx (n(i, j)/N)$. According to the definition of mutual information, the hth feature dimension is important in determining the land cover category if the value of $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ is large. As a result, we can let $p_h = I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ $(h = 1, 2, \ldots, d)$ after normalizing $I(\mathbf{X}_{\text{labeled}[:,h]}, \mathbf{Y})$ to $[0, 1]$.

It is notable that for each graph node, the spectral-level augmentation can be performed no more than once to avoid excessive damage to graph information. Meanwhile, the spectral-level augmentation will be performed on the node pairs in descending order by the value of $\mathbf{A}_{ij}$, with the expectation that feature exchange across similar nodes can probably preserve the original topological information.

V. Localized and Hierarchical Graph Convolution

In HSI, the image regions that are far away in the original 2-D space may belong to the same land cover category. However, most existing GCN models fail to exploit the long-range dependencies among image regions, as they mainly aim at encoding the pairwise importance among local image regions, which could lead to degraded performance. To deal with this issue, we propose to use localized and hierarchical graph convolution, simultaneously, so as to incorporate both local and global contextual information.

A. Localized Graph Convolution

There are various outstanding GCN architectures available to obtain representations from the local view, such as the well-known GCN [29] and graph attention network [51]. For simplicity, we select the classical GCN to perform localized graph convolution. The convolution process can be denoted as follows:

$$\mathbf{Z}^{\text{local}} = \hat{\mathbf{A}}\sigma\left(\hat{\mathbf{A}}\mathbf{X}_1\mathbf{W}^{(0)}\right)\mathbf{W}^{(1)}, \tag{7}$$

where $\hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\bar{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$ is leveraged to normalize the adjacency matrix, $\bar{\mathbf{A}} = \mathbf{A}_1 + \mathbf{I}$ is used to add the self-connection to adjacency matrix, $\mathbf{A}_1$ is the adjacency matrix of $\tilde{\mathcal{G}}_1$, $\mathbf{I}$ is the identity matrix, and $\tilde{\mathbf{D}}_{ii} = \sum_j \bar{\mathbf{A}}_{ij}$. In (7), $\mathbf{W}^{(0)}$ and $\mathbf{W}^{(1)}$ indicate the learnable weight matrices, $\sigma(\cdot)$ denotes the activation function (i.e., rectified linear unit (ReLU) [52] in our ConGCN), $\mathbf{X}_1 \in \mathbb{R}^{n\times d}$ represents the feature matrix of $\tilde{\mathcal{G}}_1$, and $\mathbf{Z}^{\text{local}}$ represents the representations learned from the local view.

B. Hierarchical Graph Convolution

To incorporate the global contextual information of HSI, we adopt the hierarchical GCN (HGCN) [53] to perform hierarchical graph convolution. Generally speaking, HGCN aggregates the nodes with similar structures to a set of hypernodes constantly. Therefore, it is able to generate coarsened graphs via successive convolution and enlarge the receptive

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field. In addition, the symmetric graph refining layers are utilized to reconstruct the original graph topology for node-level representation. By this means, the global contextual information can be gradually encoded via hierarchical graph convolution. Hence, the representations $\mathbf{Z}^{\text{global}}$ generated from the global view can well complement to $\mathbf{Z}^{\text{local}}$ and help improve the representation ability of the proposed method.

VI. Loss Function

In this section, we introduce the loss functions employed by our method.

A. Semisupervised Contrastive Loss Function

In the proposed ConGCN, we intend to leverage the abundant spectral information to help guide the model training. Fortunately, contrastive learning can naturally exploit the spectral signatures of HSI to generate representations. However, the traditional unsupervised contrastive loss [54] can only utilize unlabeled data for model training, which might ignore the precious label information of HSI. To cope with this issue, we devise a semisupervised contrastive loss to improve the discriminative power of generated representation. Specifically, it can be divided into two parts, i.e., the unsupervised and supervised contrastive losses, respectively.

Here, the unsupervised loss in the local graph view $\mathcal{L}_{\text{uc}}^{\text{local}}(\mathbf{x}_i)$ can be calculated as (8). Analogously, the unsupervised loss in the global graph view $\mathcal{L}_{\text{uc}}^{\text{global}}(\mathbf{x}_i)$ can be calculated by (9) as follows:

$$\mathcal{L}_{\text{uc}}^{\text{local}}(\mathbf{x}_i) = -\frac{1}{2n}\log\frac{e^{\left\langle \mathbf{z}_i^{\text{local}}, \mathbf{z}_i^{\text{global}} \right\rangle}}{\sum_{j=1}^{n} e^{\left\langle \mathbf{z}_i^{\text{local}}, \mathbf{z}_j^{\text{global}} \right\rangle}}, \tag{8}$$

$$\mathcal{L}_{\text{uc}}^{\text{global}}(\mathbf{x}_i) = -\frac{1}{2n}\log\frac{e^{\left\langle \mathbf{z}_i^{\text{global}}, \mathbf{z}_i^{\text{local}} \right\rangle}}{\sum_{j=1}^{n} e^{\left\langle \mathbf{z}_i^{\text{global}}, \mathbf{z}_j^{\text{local}} \right\rangle}}, \tag{9}$$

where $\mathbf{z}_i^{\text{local}} = \mathbf{Z}_{i,:}^{\text{local}}$ and $\mathbf{z}_i^{\text{global}} = \mathbf{Z}_{i,:}^{\text{global}}$ indicate the representations of $\mathbf{x}_i$ learned from local and global views, respectively, and $\langle\cdot\rangle$ expresses the inner product. The notations $\mathbf{Z}_{i,:}^{\text{local}}$ and $\mathbf{Z}_{i,:}^{\text{global}}$ denote the ith row of $\mathbf{Z}^{\text{local}}$ and $\mathbf{Z}^{\text{global}}$, respectively.

To exploit the scarce yet valuable class information for model training, the supervised contrastive loss function can be defined as follows:

$$\mathcal{L}_{\text{sc}}^{\text{local}}(\mathbf{x}_i) = -\frac{1}{2l}\log\frac{\sum_{k=1}^{l} 1_{[y_i=y_k]} e^{\left\langle \mathbf{z}_i^{\text{local}}, \mathbf{z}_k^{\text{global}} \right\rangle}}{\sum_{j=1}^{l} e^{\left\langle \mathbf{z}_i^{\text{local}}, \mathbf{z}_j^{\text{global}} \right\rangle}}, \tag{10}$$

$$\mathcal{L}_{\text{sc}}^{\text{global}}(\mathbf{x}_i) = -\frac{1}{2l}\log\frac{\sum_{k=1}^{l} 1_{[y_i=y_k]} e^{\left\langle \mathbf{z}_i^{\text{global}}, \mathbf{z}_k^{\text{local}} \right\rangle}}{\sum_{j=1}^{l} e^{\left\langle \mathbf{z}_i^{\text{global}}, \mathbf{z}_j^{\text{local}} \right\rangle}}, \tag{11}$$

where $\mathcal{L}_{\text{sc}}^{\text{local}}(\mathbf{x}_i)$ and $\mathcal{L}_{\text{sc}}^{\text{global}}(\mathbf{x}_i)$ represent the supervised pairwise contrastive losses of $\mathbf{x}_i$ in local and global views, respectively, and $\mathbb{1}_{[\cdot]}$ is an indicator function, which equals 1 if the argument inside the bracket holds and 0, otherwise. In (10) and (11), y_i and y_k are the labels of $\mathbf{x}_i$ and $\mathbf{x}_k$, respectively. Different from the unsupervised contrastive loss in (8) and (9), the positive and negative pairs in (10) and (11) can also be constructed based on the class information. That is to say, the nodes belonging to identical/different class are regarded as a positive/negative pair.

In consequence, the proposed semisupervised contrastive loss function $\mathcal{L}_{\text{ssc}}$ can be represented as follows:

$$\mathcal{L}_{\text{ssc}} = \sum_{i=1}^{n}\left(\mathcal{L}_{\text{uc}}^{\text{local}}(\mathbf{x}_i) + \mathcal{L}_{\text{uc}}^{\text{global}}(\mathbf{x}_i)\right) + \sum_{i=1}^{l}\left(\mathcal{L}_{\text{sc}}^{\text{local}}(\mathbf{x}_i) + \mathcal{L}_{\text{sc}}^{\text{global}}(\mathbf{x}_i)\right). \tag{12}$$

By minimizing $\mathcal{L}_{\text{ssc}}$, our proposed ConGCN can enhance the discriminative power of generated representations and further improve the subsequent HSI classification result.

B. Graph Generative Loss Function

In addition to the spectral information exploited by contrastive learning, we also intend to implicitly leverage spatial relations to better guide the representation learning process. Here, the graph generative loss function is designed to explore supervision signals from the spatial relations among image regions, which can lead to enhanced data representations and improved classification results.

Motivated by the generative models [55], we create a binary random variable e_{ij}, which equals 1 if there is an edge between $\mathbf{x}_i$ and $\mathbf{x}_j$, and 0, otherwise. We presume that e_{ij} is conditionally independent, so given $\mathbf{Z}^{\text{local}}$ and $\mathbf{Z}^{\text{global}}$, the conditional probability of the input graph $\mathcal{G}$ can be expressed by maximizing the following likelihood estimation:

$$p\left(\mathcal{G} \mid \mathbf{Z}^{\text{local}}, \mathbf{Z}^{\text{global}}\right) = \prod_{i,j} p\left(e_{ij} \mid \mathbf{Z}^{\text{local}}, \mathbf{Z}^{\text{global}}\right). \tag{13}$$

Furthermore, we reasonably assume that the probability of e_{ij} only depends on the representations of $\mathbf{x}_i$ and $\mathbf{x}_j$ according to [55]. As a result, the conditional probability of e_{ij} can be computed as $p\left(e_{ij} \mid \mathbf{Z}^{\text{local}}, \mathbf{Z}^{\text{global}}\right) = p\left(e_{ij} \mid \mathbf{z}_i^{\text{local}}, \mathbf{z}_j^{\text{global}}\right)$.

In the end, the logistic function is utilized to encode the abovementioned conditional probability, which turns out to be

$$\begin{aligned} p\left(\mathcal{G} \mid \mathbf{Z}^{\text{local}}, \mathbf{Z}^{\text{global}}\right) &= \prod_{i,j} p\left(e_{ij} \mid \mathbf{z}_i^{\text{local}}, \mathbf{z}_j^{\text{global}}\right) \\ &= \prod_{i,j} \delta\left(\left[\mathbf{z}_i^{\text{local}}, \mathbf{z}_j^{\text{global}}\right]\mathbf{w}\right), \end{aligned} \tag{14}$$

where $\delta(\cdot)$ denotes the logistic function, $\mathbf{w}$ represents a trainable parameter vector, and $[\cdot,\cdot]$ denotes the concatenation operation. Consequently, the proposed graph generative loss function can be represented as $\mathcal{L}_{\text{g}^2} = -p\left(\mathcal{G} \mid \mathbf{Z}^{\text{local}}, \mathbf{Z}^{\text{global}}\right)$, which is used by our proposed ConGCN to extract precious yet implicit spatial relations among image regions in HSI.

C. Model Training

After integrating the graph representation from global and local graph views, the final output of our proposed ConGCN can be computed as $\mathbf{O} = \lambda_{\text{local}}\mathbf{Z}^{\text{local}} + (1-\lambda_{\text{local}})\mathbf{Z}^{\text{global}}$, where $0 < \lambda_{\text{local}} < 1$ denotes the weight assigned to $\mathbf{Z}^{\text{local}}$. In addition, the cross-entropy loss function $\mathcal{L}_{\text{ce}} = -\sum_{i=1}^{l}\sum_{j=1}^{c}\mathbf{Y}_{ij}\ln\mathbf{O}_{ij}$ is utilized to penalize the label differences between the final output $\mathbf{O}$ and the initially labeled seed superpixels.

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Algorithm 1 Proposed ConGCN Algorithm

Input: Feature matrix $\mathbf{X}$; label matrix $\mathbf{Y}$; maximum number of iterations $\mathcal{T}$.

Output: Predicted label for each unlabeled graph node.

for $t = 1$ to $\mathcal{T}$ **do**
  Generate two augmented graphs $\tilde{\mathcal{G}}_1$ and $\tilde{\mathcal{G}}_2$;
  Perform localized and hierarchical graph convolution to obtain graph representation $\mathbf{Z}^{\text{local}}$ and $\mathbf{Z}^{\text{global}}$, respectively;
  Compute semi-supervised contrastive loss function $\mathcal{L}_{\text{ssc}}$ via Eq. (12);
  Compute graph generative loss function $\mathcal{L}_{\text{g}^2}$;
  Compute cross-entropy loss $\mathcal{L}_{\text{ce}}$;
  Update network parameters via operating back propagation according to overall loss function $\mathcal{L}$ in Eq. (15);
  $t := t + 1$;
end for
Predict labels based on the trained network.

Finally, the overall loss function of our proposed ConGCN is shown in (15) via assembling the cross-entropy loss function $\mathcal{L}_{\text{ce}}$, semisupervised contrastive loss function $\mathcal{L}_{\text{ssc}}$, and graph generative loss function $\mathcal{L}_{\text{g}^2}$, namely

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_{\text{ssc}}\mathcal{L}_{\text{ssc}} + \lambda_{\text{g}^2}\mathcal{L}_{\text{g}^2}, \tag{15}$$

where $\lambda_{\text{ssc}} > 0$ and $\lambda_{\text{g}^2} > 0$ are hyperparameters adjusting the impact of $\mathcal{L}_{\text{ssc}}$ and $\mathcal{L}_{\text{g}^2}$, respectively. The process of the proposed ConGCN is exhibited in Algorithm 1.

VII. Experiments

To demonstrate the effectiveness of our proposed ConGCN, intensive experiments are conducted on four well-known HSI datasets, namely, *Indian Pines*, *University of Pavia*, *Salinas*, and *Houston University* datasets, of which the details are introduced in Section I of the Appendix. Concretely, we first compare our proposed ConGCN with other state-of-the-art methods using four metrics according to [1], [2], and [3], namely, per-class accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient. After that, ablative experiments are carried out to evaluate the effectiveness of semisupervised contrastive loss function, graph generative loss function, and our proposed graph augmentation technique. To further evaluate the effectiveness of our proposed method, we conduct experiments on two additional datasets: *WHU-Hi-HanChuan* and *WHU-Hi-HongHu* datasets [56], [57], which are presented in Section II-B of the Appendix. Meanwhile, to reveal the advantage of our proposed ConGCN over the baseline methods in terms of efficiency, we report the training and inferring time of the proposed ConGCN and the compared baseline methods, which are presented in Section II-D of the Appendix.

A. Experimental Settings

In our experiments, the proposed ConGCN is implemented by TensorFlow with Adam optimizer. For each dataset, 30 labeled pixels (i.e., examples) of HSI are randomly selected in each land cover category for training. If there are less than 30 examples, only 15 labeled examples are chosen for the corresponding land cover category. In the training phase, 90% of the labeled examples are utilized to train the network parameters, and 10% of the labeled examples are leveraged as validation set to fine-tune the hyperparameters (e.g., λ_{local} and λ_{ssc}). All unlabeled examples are used for testing to evaluate the classification performance. The network architecture of our proposed ConGCN is kept identical for all datasets. Moreover, the learning rate and the maximum number of iterations are set to 0.01 and 4000, respectively.

To demonstrate the effectiveness of our proposed ConGCN method, other state-of-the-art HSI classification methods are also utilized for comparison. Concretely, we adopt three GCN-based methods, namely, dual-level deep spatial manifold representation (SMR) network [58], multilevel superpixel structured graph U-net (MSSGU) [33], and superpixel graph learning (SGL) [59]. Besides, one contrastive learning-based method [42] termed "self-supervised contrastive learning" (SSCL) is employed for comparison. In addition, we incorporate two CNN-based methods, namely, attention-based adaptive spectral–spatial kernel (A^2S^2K) [24] and adaptive spectral–spatial multiscale network (ASSMN) [25]. The compared baseline methods also include one generative-based method, i.e., adaptive dropblock-enhanced GAN (ADGAN) [22]. Besides, the proposed ConGCN is compared with four traditional HSI classification methods, namely, multiple feature learning (MFL) [60], joint collaborative representation and SVM with decision fusion (JSDF) [61], probabilistic-kernel collaborative representation classification (PKCRC) [16], and image fusion and recursive filtering (IFRF) [17]. All methods are repeated ten times, where the mean accuracies and standard deviations are also reported.

B. Experimental Results

To evaluate the performance of our proposed ConGCN, ConGCN is compared with the abovementioned baseline methods in both quantitative and qualitative aspects.

1) Results on the Indian Pines Dataset: The quantitative results obtained by different methods on the *Indian Pines* dataset are summarized in Table II. We observe that the proposed ConGCN achieves the top-level performance among all the methods in terms of OA and kappa coefficient. The standard deviations are relatively small as well. Note that the proposed method achieves 100% accuracies on five land cover categories (i.e., IDs = 4, 7, 8, 9, and 13) and can generally acquire stable and high classification accuracies on the remaining categories. Therefore, it is reasonable to infer that the proposed ConGCN is more stable and effective than the compared methods. The improvements of ConGCN over both CNN-based methods (i.e., A^2S^2K and ASSMN) are even higher than 8% in terms of OA, which indicates the great potential of GCN in HSI classification. Moreover, it is notable that the ConGCN performs better than three GCN-based methods (i.e., SMR, MSSGU, and SGL), where the effectiveness of contextual information at both local and global levels is validated. The AA of our ConGCN is slightly lower than that of MSSGU, as MSSGU can adaptively

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TABLE II

PER-CLASS ACCURACIES, OAS, AAS (%), AND KAPPA COEFFICIENTS ACHIEVED BY DIFFERENT METHODS ON THE *Indian Pines* DATASET. THE BEST AND SECOND BEST RECORDS IN EACH ROW ARE **BOLDED** AND UNDERLINED, RESPECTIVELY

ID	SMR [58]	MSSGU [33]	SGL [59]	SSCL [42]	A^2S^2K [24]	ASSMN [25]	ADGAN [22]	MFL [60]	JSDF [61]	PKCRC [16]	IFRF [17]	ConGCN
1	98.91±3.26	**100.00±0.00**	**100.00±0.00**	00.00±0.00	79.31±18.46	**100.00±0.00**	**100.00±0.00**	98.06±0.58	**100.00±0.00**	**100.00±0.00**	77.49±35.90	98.75±3.75
2	63.81±13.64	91.69±1.52	89.08±4.56	43.62±4.44	87.85±4.01	78.83±3.79	68.59±10.54	74.72±0.66	90.75±3.19	75.49±5.15	**93.25±3.40**	92.07±1.55
3	69.32±15.02	**98.35±0.60**	90.00±3.24	32.84±9.89	86.00±6.73	83.78±6.61	58.80±11.33	82.14±0.70	77.84±3.81	78.91±13.08	84.41±9.04	97.50±0.64
4	92.36±11.51	98.13±0.56	97.10±4.43	0.93±2.00	86.14±7.12	94.49±1.15	95.02±2.98	93.60±0.55	99.86±0.33	**100.00±0.00**	84.68±9.64	**100.00±0.00**
5	82.88±9.25	95.92±0.81	**97.75±1.73**	29.15±17.33	91.95±2.83	90.84±1.57	75.74±9.50	92.54±0.43	87.20±2.73	91.39±4.27	94.42±6.21	94.50±1.79
6	87.77±6.99	**99.84±0.23**	99.30±0.42	69.05±9.90	95.36±1.53	90.56±1.75	91.54±9.22	98.40±0.27	98.54±0.28	99.34±0.78	96.85±2.18	98.99±0.04
7	97.14±8.57	**100.00±0.00**	0.00±0.00	0.00±0.00	47.39±9.54	**100.00±0.00**	**100.00±0.00**	97.28±0.45	**100.00±0.00**	96.92±3.77	62.99±29.26	**100.00±0.00**
8	95.29±8.04	**100.00±0.00**	**100.00±0.00**	88.28±13.10	99.88±0.28	**100.00±0.00**	94.84±13.74	99.82±0.05	99.80±0.31	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	98.50±4.50	**100.00±0.00**	0.00±0.00	0.00±0.00	30.17±5.98	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	54.22±23.22	**100.00±0.00**
10	82.90±12.13	**96.26±1.52**	90.50±6.12	48.85±14.17	67.29±7.01	89.68±3.23	82.00±5.35	84.59±0.53	89.99±4.24	88.20±5.64	87.84±5.35	93.57±2.86
11	59.46±10.33	91.54±0.78	94.76±3.50	77.03±4.96	**97.21±1.51**	76.62±3.82	65.56±9.75	83.73±0.39	76.75±5.12	79.97±3.64	95.53±2.00	97.12±1.19
12	79.45±16.60	**98.47±0.51**	94.33±2.67	37.67±13.70	83.37±8.30	91.72±3.10	83.53±7.38	83.68±0.72	87.10±2.82	93.80±1.93	89.44±5.78	97.57±0.87
13	94.63±6.37	**100.00±0.00**	99.09±0.38	47.02±39.63	85.05±4.42	**100.00±0.00**	**100.00±0.00**	99.20±0.06	99.89±0.36	**100.00±0.00**	91.99±4.87	**100.00±0.00**
14	93.91±4.40	**99.98±0.04**	99.84±0.24	78.35±8.32	99.43±0.35	94.01±2.02	91.17±6.65	96.80±0.40	97.21±2.78	93.30±2.29	99.30±0.63	99.83±0.02
15	62.62±15.17	**99.77±0.36**	99.58±0.42	12.56±11.33	87.97±5.28	98.46±1.19	92.89±0.99	97.86±0.20	99.58±0.68	95.62±5.26	92.33±4.26	99.41±0.08
16	96.67±6.84	**100.00±0.00**	**100.00±0.00**	1.94±3.81	85.43±4.50	**100.00±0.00**	99.05±1.05	98.72±0.35	**100.00±0.00**	**100.00±0.00**	96.59±1.32	87.14±2.18
OA	75.84±9.40	95.87±0.11	94.35±0.93	55.53±1.63	88.31±1.51	86.38±1.78	77.51±5.53	87.38±0.12	88.34±1.39	86.88±1.57	92.56±1.55	**96.74±0.50**
AA	84.73±8.30	**98.12±0.09**	84.46±0.42	35.46±3.18	81.86±1.38	93.06±1.00	87.42±3.36	92.57±0.10	94.03±0.55	93.31±1.25	87.58±3.02	97.28±0.29
kappa	72.87±10.33	95.28±0.13	93.53±1.06	48.47±2.03	86.76±1.67	84.51±2.02	74.98±5.93	85.64±0.14	86.80±1.55	85.09±1.81	91.51±1.74	**96.27±0.57**

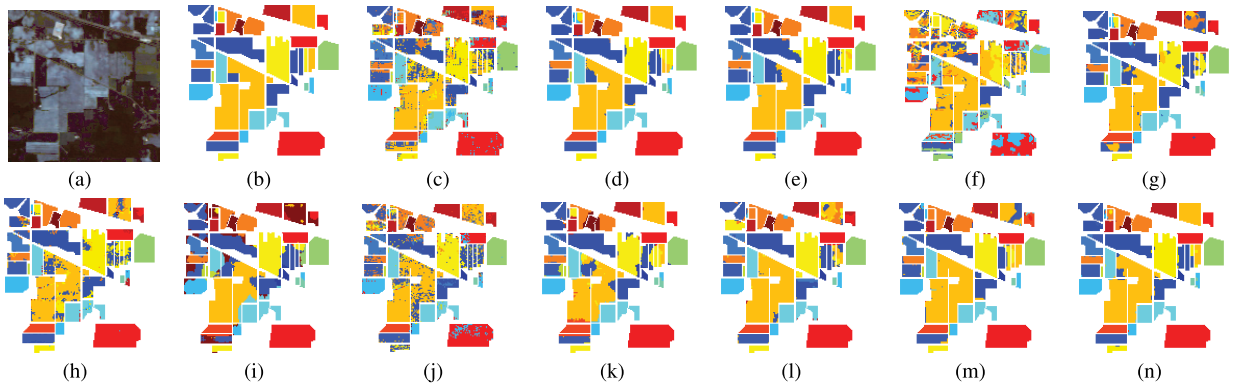

Fig. 4. Classification maps obtained by different methods on the *Indian Pines* dataset. (a) False-color image. (b) Ground-truth map. (c) SMR. (d) MSSGU. (e) SGL. (f) SSCL. (g) A^2S^2K. (h) ASSMN. (i) ADGAN. (j) MFL. (k) JSDF. (l) PKCRC. (m) IFRF. (n) ConGCN (proposed).

TABLE III

PER-CLASS ACCURACIES, OAS, AAS (%), AND KAPPA COEFFICIENTS ACHIEVED BY DIFFERENT METHODS ON THE *University of Pavia* DATASET. THE BEST AND SECOND BEST RECORDS IN EACH ROW ARE **BOLDED** AND UNDERLINED, RESPECTIVELY

ID	SMR [58]	MSSGU [33]	SGL [59]	SSCL [42]	A^2S^2K [24]	ASSMN [25]	ADGAN [22]	MFL [60]	JSDF [61]	PKCRC [16]	IFRF [17]	ConGCN
1	96.96±1.89	**98.14±0.72**	88.26±3.06	96.94±1.76	94.25±2.75	78.33±3.30	49.97±20.68	94.45±0.25	82.40±4.07	96.69±3.22	85.15±7.06	93.11±2.43
2	82.00±6.11	82.90±2.30	97.24±1.45	97.90±1.90	97.92±2.08	74.97±1.80	33.82±23.16	90.17±0.65	90.76±3.74	92.57±7.68	**98.83±0.53**	96.55±1.87
3	95.21±2.60	**100.00±0.00**	94.19±2.64	14.89±16.14	72.27±11.87	81.14±6.84	70.07±35.81	85.05±0.54	86.71±4.14	94.07±6.85	83.46±4.72	97.24±1.83
4	94.73±3.47	**98.44±0.45**	92.76±1.97	42.87±19.21	78.73±5.59	97.76±1.06	54.79±24.08	93.31±0.28	92.88±2.16	91.63±4.96	85.76±7.41	93.91±0.25
5	99.03±1.48	**100.00±0.00**	99.30±0.82	87.00±13.48	99.92±0.02	99.98±0.03	91.10±25.17	99.38±0.02	**100.00±0.00**	**100.00±0.00**	99.50±0.74	98.80±0.15
6	99.79±0.36	**100.00±0.00**	99.80±0.51	9.43±7.15	81.74±7.58	90.76±3.69	58.62±31.24	93.31±0.20	94.30±4.55	99.38±1.04	91.21±3.12	**100.00±0.00**
7	97.59±5.41	**100.00±0.00**	99.31±0.23	14.21±12.56	81.40±9.44	98.43±1.26	85.29±29.34	99.39±0.04	96.62±1.37	**100.00±0.00**	84.24±6.12	99.12±0.44
8	88.79±14.78	**99.48±0.97**	94.07±2.12	81.36±14.61	70.44±6.86	89.38±1.92	48.79±31.33	85.30±0.53	94.69±3.74	98.79±1.60	74.96±4.86	94.76±1.93
9	99.09±1.47	99.86±0.11	99.67±0.00	2.30±6.87	95.60±4.33	99.87±0.18	90.08±7.88	99.76±0.02	99.56±0.36	**99.89±0.00**	66.16±12.01	82.81±3.20
OA	90.13±3.94	92.06±1.01	95.58±0.60	72.85±2.44	87.82±1.47	82.54±0.99	48.35±17.95	91.54±0.30	90.82±1.30	95.16±3.68	90.10±1.57	**95.97±0.90**
AA	94.80±3.11	**97.65±0.26**	96.07±0.47	49.66±3.58	85.81±1.82	90.07±1.20	64.73±16.71	93.35±0.10	93.10±0.65	97.00±1.28	85.48±1.86	95.14±0.57
kappa	87.51±4.86	89.81±1.26	94.17±0.78	61.18±3.55	84.32±1.88	77.87±1.20	42.94±17.63	88.98±0.38	88.02±1.62	93.73±4.69	87.04±2.03	**94.69±1.17**

incorporate features for various land cover categories from the graphs at different levels. Hence, MSSGU obtains better performance than ConGCN in several land cover categories, such as "Soybean-notill" (ID = 10) and "Stone-steel-towers" (ID = 16). Due to the instability of GAN during training [62], ADGAN has large standard deviations. Consequently, the four traditional HSI classification methods (namely, MFL, JSDF, PKCRC, and IFRF) achieve higher OAs, AAs, and kappa coefficients than ADGAN, indicating the effectiveness and stability of conventional pattern recognition methods. Although the SSCL employs contrastive learning, its classification performance can still be unsatisfactory, especially in the land cover categories with IDs = 1, 4, 7, 9, 15, and 16. This is because SSCL only utilizes unlabeled examples to pretrain the encoder and simply fine-tunes the network with few labeled examples. Different from SSCL, our proposed ConGCN utilizes the unlabeled data and the available class information, simultaneously, for model training. As a result, our proposed method outperforms SSCL on the *Indian Pines* dataset.

Fig. 4 shows a visual comparison of the classification maps generated by different methods on the *Indian Pines* dataset, where the ground-truth map is exhibited in Fig. 4(b). The classification maps obtained by SMR [Fig. 4(c)], ASSMN [Fig. 4(h)], and MFL [Fig. 4(j)] suffer from pepper-noise-like mistakes within multiple areas. For instance, the middle parts of the classification maps obtained by SMR [Fig. 4(c)], ASSMN [Fig. 4(h)], and MFL [Fig. 4(j)] are highly confusing. Comparatively, the classification map of the proposed ConGCN method yields a smoother visual effect and shows fewer misclassifications than the compared methods.

2) Results on the University of Pavia Dataset: Table III presents the quantitative results of different methods on the *University of Pavia* dataset. Similar to the results on the

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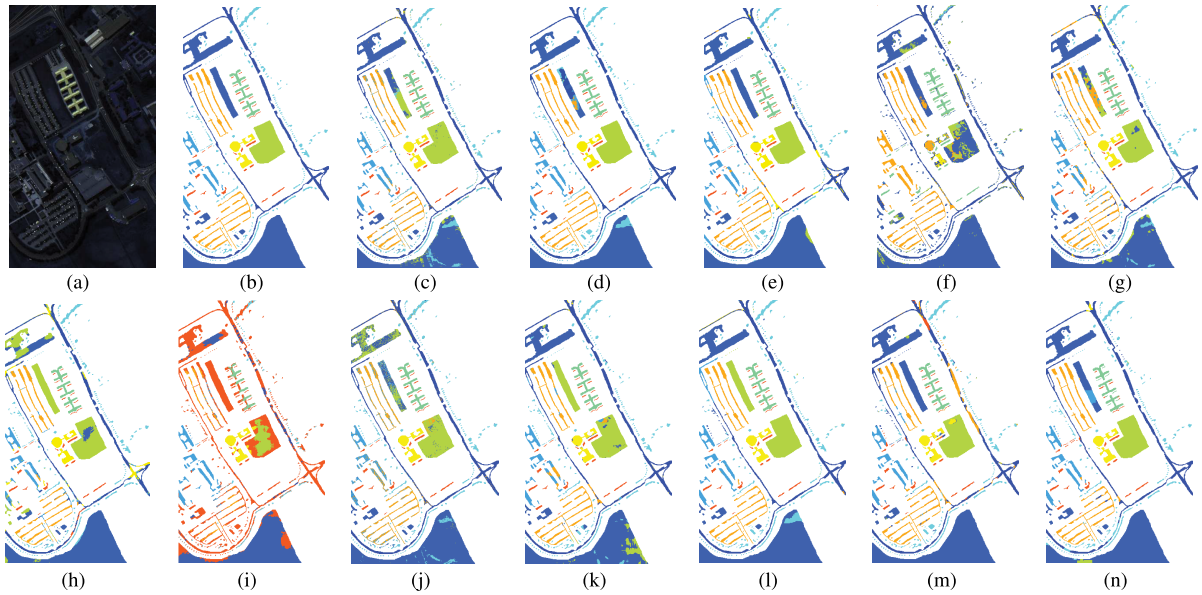

Fig. 5. Classification maps obtained by different methods on the *University of Pavia* dataset. (a) False-color image. (b) Ground-truth map. (c) SMR. (d) MSSGU. (e) SGL. (f) SSCL. (g) A^2S^2K. (h) ASSMN. (i) ADGAN. (j) MFL. (k) JSDF. (l) PKCRC. (m) IFRF. (n) ConGCN (proposed).

TABLE IV

PER-CLASS ACCURACIES, OAS, AAS (%), AND KAPPA COEFFICIENTS ACHIEVED BY DIFFERENT METHODS ON THE *Salinas* DATASET. THE BEST AND SECOND BEST RECORDS IN EACH ROW ARE **BOLDED** AND UNDERLINED, RESPECTIVELY

ID	SMR [58]	MSSGU [33]	SGL [59]	SSCL [42]	A^2S^2K [24]	ASSMN [25]	ADGAN [22]	MFL [60]	JSDF [61]	PKCRC [16]	IFRF [17]	ConGCN
1	99.82±0.31	**100.00±0.00**	**100.00±0.00**	81.57±13.21	**100.00±0.00**	97.68±6.17	95.36±6.40	99.63±0.07	**100.00±0.00**	**100.00±0.00**	99.98±0.06	**100.00±0.00**
2	99.53±0.73	99.99±0.03	**100.00±0.00**	81.56±8.56	99.98±0.05	99.02±0.46	54.19±42.34	99.34±0.06	**100.00±0.00**	**100.00±0.01**	**100.00±0.00**	**100.00±0.00**
3	99.74±0.21	**100.00±0.00**	**100.00±0.00**	43.48±10.02	99.98±0.02	99.57±0.39	85.70±20.43	99.77±0.03	**100.00±0.00**	**100.00±0.00**	99.79±0.26	**100.00±0.00**
4	99.86±0.28	99.88±0.07	98.15±0.85	91.79±14.35	99.25±0.42	99.87±0.14	95.09±2.96	98.88±0.07	99.93±0.09	**99.94±0.03**	96.89±1.44	98.50±0.82
5	97.23±1.18	99.50±0.17	98.36±0.05	95.72±1.36	99.52±0.28	98.62±0.86	92.34±14.98	98.72±0.04	**99.77±0.31**	99.20±0.16	99.68±0.44	97.58±0.60
6	99.53±1.04	**100.00±0.00**	**100.00±0.00**	98.00±2.05	99.99±0.02	99.93±0.10	83.01±28.61	99.18±0.11	**100.00±0.00**	99.99±0.01	99.99±0.02	99.81±0.07
7	99.81±0.16	**100.00±0.01**	99.89±0.00	99.26±0.59	99.95±0.08	99.81±0.10	96.40±3.93	98.61±0.12	99.99±0.01	**100.00±0.00**	99.66±0.16	99.94±0.01
8	86.35±7.29	97.65±0.70	98.52±0.39	79.63±5.01	88.89±2.12	77.54±7.53	35.07±36.66	76.57±0.71	87.79±4.89	85.09±7.50	**99.06±1.39**	98.33±1.16
9	99.37±0.63	**100.00±0.00**	**100.00±0.00**	97.57±2.33	99.49±0.05	99.06±0.38	92.68±16.75	99.01±0.05	99.67±0.33	**100.00±0.00**	99.95±0.13	**100.00±0.00**
10	97.19±1.20	99.47±0.10	97.98±1.70	66.90±15.77	98.74±0.68	97.04±1.87	96.82±2.36	93.10±0.30	96.53±2.55	97.75±0.74	**99.53±0.62**	99.28±0.68
11	97.05±1.93	**100.00±0.00**	97.86±2.27	40.83±36.55	**100.00±0.00**	98.85±1.40	99.03±0.03	96.81±0.30	99.76±0.21	98.15±1.83	99.17±0.30	99.74±0.09
12	**100.00±0.00**	99.87±0.17	99.74±0.00	86.39±15.94	99.93±0.13	99.76±0.38	93.63±3.82	98.84±0.20	**100.00±0.00**	**100.00±0.00**	99.66±0.36	98.26±0.71
13	99.93±0.17	**100.00±0.00**	98.70±0.06	28.77±35.22	99.95±0.08	99.40±0.26	95.49±3.60	99.38±0.08	**100.00±0.00**	98.89±0.45	97.78±1.40	97.58±0.47
14	98.91±0.69	99.81±0.23	94.86±1.09	68.89±26.04	**99.95±0.07**	98.31±1.32	98.41±1.21	96.20±0.32	98.71±0.72	99.50±0.63	96.98±1.92	98.86±0.46
15	77.41±9.95	**99.95±0.07**	98.96±0.36	61.92±10.33	85.44±4.44	89.34±7.49	94.48±4.49	78.85±0.56	81.86±5.26	85.31±10.71	94.27±2.92	99.63±0.29
16	99.30±0.51	**100.00±0.00**	98.62±0.90	42.50±23.01	99.57±0.67	97.87±0.59	96.11±0.52	99.69±0.06	98.99±0.63	99.44±0.38	99.87±0.19	**100.00±0.00**
OA	93.60±0.93	**99.43±0.14**	99.08±0.11	78.34±3.52	95.44±0.54	93.17±1.56	78.31±9.20	91.20±0.13	94.67±0.77	94.63±1.74	98.69±0.50	99.25±0.29
AA	96.94±0.37	**99.76±0.05**	98.85±0.18	72.80±5.02	98.17±0.22	96.98±0.94	87.74±4.92	95.79±0.04	97.69±0.34	97.70±0.64	98.89±0.25	99.22±0.19
kappa	92.87±1.03	**99.37±0.16**	98.97±0.12	75.78±3.97	94.91±0.60	92.41±1.73	76.52±9.84	90.21±0.14	94.06±0.85	94.03±1.94	98.54±0.55	99.17±0.33

Indian Pines dataset, in Table III, ConGCN outperforms the compared methods in terms of OA and kappa coefficient, which again validates the strength of our proposed contrastive learning-based graph convolution. Besides, the GCN-based methods (i.e., SMR, MSSGU, and SGL) obtain higher OAs, AAs, and kappa coefficients than CNN-based methods (i.e., A^2S^2K and ASSMN), which indicates the great potential of GCN in HSI classification. It is also notable that the performance of SMR on the *University of Pavia* dataset is better than that on the *Indian Pines* dataset. Since SMR can flexibly capture the variations around irregular boundaries with different convolutional kernels, the advantage of SMR becomes prominent on datasets containing various irregular boundaries, such as the *University of Pavia* dataset. Moreover, compared with the CNN-based methods, the proposed ConGCN outperforms the CNN-based methods (i.e., A^2S^2K and ASSMN) with substantial margins (i.e., 8.15% and 13.43%), which suggests that both spectral signatures and spatial relations captured by our ConGCN are more useful than the information characterized by the fixed convolutional kernels of CNN. The OAs of four traditional methods can reach 90% or higher, which confirms the good discriminability of conventional pattern recognition methods.

Fig. 5 visualizes the classification results generated by different methods on the *University of Pavia* dataset. As depicted in Fig. 5(n), the classification map of our proposed ConGCN is noticeably closer to the ground-truth map [Fig. 5(b)] than those of other methods, which is consistent with previous results in Table III. Besides, A^2S^2K [Fig. 5(g)] and ASSMN [Fig. 5(h)], which use the fixed convolutional kernels, produce more errors than ConGCN. In the classification map of ADGAN [Fig. 5(i)], most of the pixels are misclassified as the same land cover category, which illustrates the reason for ADGAN's poor OA. It is also notable that large numbers of pixels in the classification map of SSCL [Fig. 5(f)] are misclassified. It indicates that the performance of simply using the traditional paradigm of contrastive learning is far from satisfactory, especially with few labeled examples.

3) Results on the Salinas Dataset: The experimental results of different methods on the *Salinas* dataset are presented in Table IV. It is apparent that the performances of nearly all the methods on the *Salinas* dataset are better than those on the *Indian Pines* dataset and the *University of Pavia* dataset. The reason might be that the boundaries in the *Salinas* dataset are more regular than those in the *Indian Pines* dataset and the *University of Pavia* dataset. As a result, the regions of the

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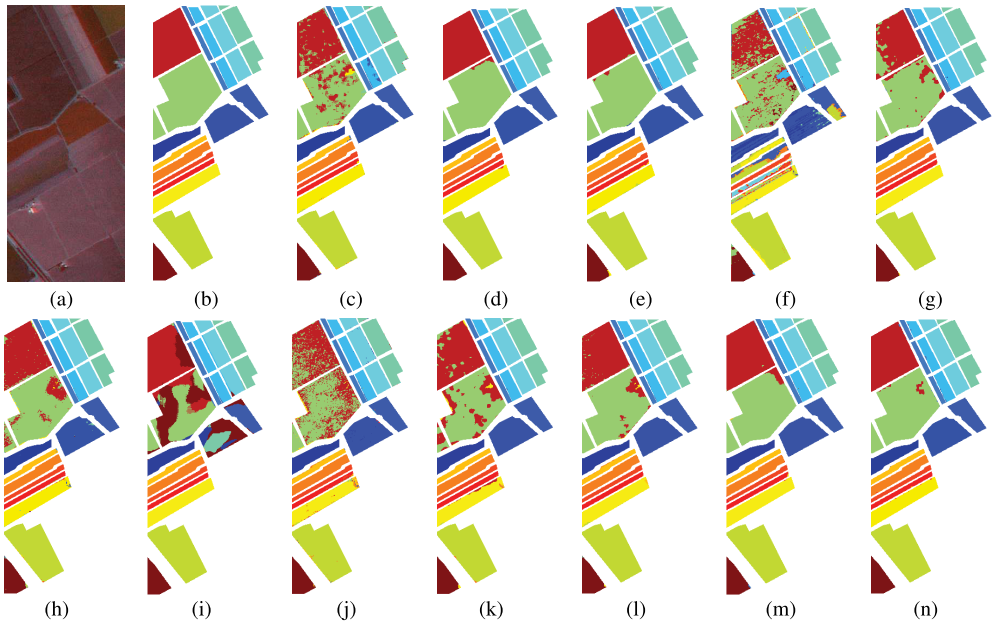

Fig. 6. Classification maps obtained by different methods on the *Salinas* dataset. (a) False-color image. (b) Ground-truth map. (c) SMR. (d) MSSGU. (e) SGL. (f) SSCL. (g) A^2S^2K. (h) ASSMN. (i) ADGAN. (j) MFL. (k) JSDF. (l) PKCRC. (m) IFRF. (n) ConGCN (proposed).

TABLE V

PER-CLASS ACCURACIES, OAS, AAS (%), AND KAPPA COEFFICIENTS ACHIEVED BY DIFFERENT METHODS ON THE *Houston University* DATASET. THE BEST AND SECOND BEST RECORDS IN EACH ROW ARE **BOLDED** AND UNDERLINED, RESPECTIVELY

ID	SMR [58]	MSSGU [33]	SGL [59]	SSCL [42]	A^2S^2K [24]	ASSMN [25]	ADGAN [22]	MFL [60]	JSDF [61]	PKCRC [16]	IFRF [17]	ConGCN
1	92.64±3.13	**98.35±2.46**	89.13±6.46	78.18±8.63	97.80±1.49	94.52±2.79	60.27±24.21	91.00±0.90	97.41±1.21	79.00±1.38	81.73±6.26	97.79±0.92
2	98.21±1.01	91.81±2.24	75.79±6.67	42.96±11.79	98.53±1.02	97.46±2.53	23.69±16.97	94.97±0.62	**99.48±0.25**	75.80±1.37	74.54±4.78	97.35±0.59
3	97.14±1.97	99.60±0.34	99.60±0.04	95.91±1.20	99.76±0.48	98.22±0.29	94.49±6.77	99.74±0.01	99.88±0.22	**100.00±0.00**	93.65±2.34	98.25±0.32
4	98.28±1.78	99.68±0.48	75.16±2.67	66.31±7.73	97.00±1.08	92.75±7.85	42.58±16.45	93.14±0.45	98.22±2.80	82.89±1.77	70.83±5.22	**99.74±0.52**
5	97.75±1.01	**100.00±0.00**	99.05±0.47	70.72±13.86	98.55±0.94	97.08±1.26	53.16±16.55	98.36±0.15	**100.00±0.00**	88.53±2.76	92.96±2.28	99.79±0.33
6	98.58±1.23	**99.63±0.65**	96.47±2.54	10.62±15.89	98.70±1.81	98.07±1.06	91.83±7.87	97.24±0.40	99.32±1.09	79.86±3.54	74.66±10.71	97.29±0.00
7	92.93±2.78	95.33±1.36	69.51±6.61	64.06±10.15	94.74±3.38	86.52±5.93	57.26±33.71	88.02±0.52	91.93±4.91	85.85±4.41	81.42±2.77	**98.18±0.43**
8	81.22±3.26	90.30±2.88	67.83±5.24	19.08±10.07	95.53±5.09	73.38±3.93	31.79±11.21	64.28±0.71	68.82±6.16	51.28±8.90	79.36±7.30	**97.75±1.38**
9	85.01±4.59	90.66±3.00	75.73±4.31	67.63±10.58	92.19±5.37	72.18±7.57	26.59±15.06	67.91±0.57	69.47±8.56	86.34±2.44	90.30±4.09	**97.77±1.28**
10	90.37±5.50	**100.00±0.00**	97.14±7.22	29.60±15.16	83.72±2.76	82.47±5.35	27.29±23.47	87.64±0.95	85.63±9.32	70.90±19.42	86.09±3.79	**100.00±0.00**
11	91.74±7.04	**99.99±0.02**	94.91±2.77	50.08±7.59	90.78±4.37	82.91±6.05	48.64±24.83	89.20±0.47	94.51±3.82	65.52±2.72	89.99±5.18	98.91±0.96
12	85.85±5.03	92.78±2.08	79.31±5.19	21.69±9.27	91.31±6.98	76.80±7.85	73.49±27.32	77.65±0.46	84.33±5.33	52.08±6.43	91.08±3.70	**99.45±1.28**
13	83.37±7.43	86.76±2.48	92.31±2.84	19.68±14.86	96.59±3.49	70.46±8.76	68.81±17.97	80.76±0.40	98.10±1.28	78.81±5.95	75.39±10.54	**98.69±2.00**
14	96.21±8.18	**100.00±0.00**	98.45±1.02	69.96±12.94	**100.00±0.00**	98.94±1.07	90.29±8.00	98.01±0.29	**100.00±0.00**	**100.00±0.00**	88.34±3.31	99.38±1.86
15	99.29±0.85	**99.99±0.04**	96.81±0.45	74.38±11.62	99.13±0.72	98.70±0.96	93.60±6.11	98.47±0.11	99.86±0.36	97.90±0.31	91.22±4.15	98.19±0.32
OA	91.89±1.93	96.03±0.52	84.71±1.28	52.69±2.23	94.37±0.74	86.60±3.33	51.57±9.80	86.66±0.13	90.51±0.95	76.98±1.41	83.96±1.11	**98.61±0.37**
AA	92.57±1.96	96.33±0.45	87.15±1.04	52.06±2.82	95.62±0.46	88.03±3.02	58.92±8.97	88.43±0.11	92.46±0.75	79.65±1.31	84.10±1.34	**98.57±0.41**
kappa	91.22±2.09	95.71±0.56	83.49±1.38	48.86±2.40	93.90±0.80	85.50±3.60	48.34±10.36	85.56±0.14	89.74±1.03	75.07±1.53	82.66±1.20	**98.49±0.40**

Salinas dataset are more distinguishable than those of the *Indian Pines* dataset and the *University of Pavia* dataset. Although the proposed ConGCN is in the second place, it is only 0.18% lower than MSSGU in terms of OA. It is noticeable that MSSGU, SGL, and our proposed ConGCN all achieve good classification results, outperforming the CNN-based baseline methods (i.e., A^2S^2K and ASSMN) regarding OA, AA, and kappa coefficient, respectively. It verifies the effectiveness of GCN-based methods in HSI classification. It is noted that the OA of IFRF is slightly lower than that of ConGCN. The IFRF is a filter-based method, which utilizes recursive filtering to explore spatial information. Different from IFRF, our proposed method jointly incorporates contextual information at different spatial levels by employing localized and hierarchical graph convolution, simultaneously, thus revealing better representation ability than IFRF.

Fig. 6 visualizes the classification results generated by different methods. The classification maps obtained by SMR [Fig. 6(c)], SSCL [Fig. 6(f)], ASSMN [Fig. 6(h)], and MFL [Fig. 6(j)] suffer from noise-like mistakes within multiple areas. On the contrary, it can be observed that the classification map of our proposed ConGCN [Fig. 6(n)] is less noisy than those of other methods, which is consistent with the results listed in Table IV.

4) Results on the Houston University Dataset: Table V shows the classification results obtained by different methods on the *Houston University* dataset. We can observe that the proposed ConGCN outperforms all the compared methods by a substantial margin in terms of OA, AA, and kappa coefficient. In addition, the standard deviations of ConGCN are relatively small as well. Another notable fact is that our proposed ConGCN outperforms other methods in seven land cover categories, which reveals the effectiveness of our method. Furthermore, our proposed ConGCN obtains OA increments of 6.72%, 2.58%, and 13.90% in comparison with the three GCN-based methods, namely, SMR, MSSGU, and SGL, respectively. It can be inferred that the semisupervised contrastive loss and the graph generative loss designed by our method are beneficial to improve the discriminative ability of the GCN-based methods. Meanwhile, we can see that although A^2S^2K captures discriminative spectral–spatial features with the spectral attention mechanism, its OA is still lower than that of ConGCN. We speculate that the supervision signals extracted from the spectral–spatial information of HSI are

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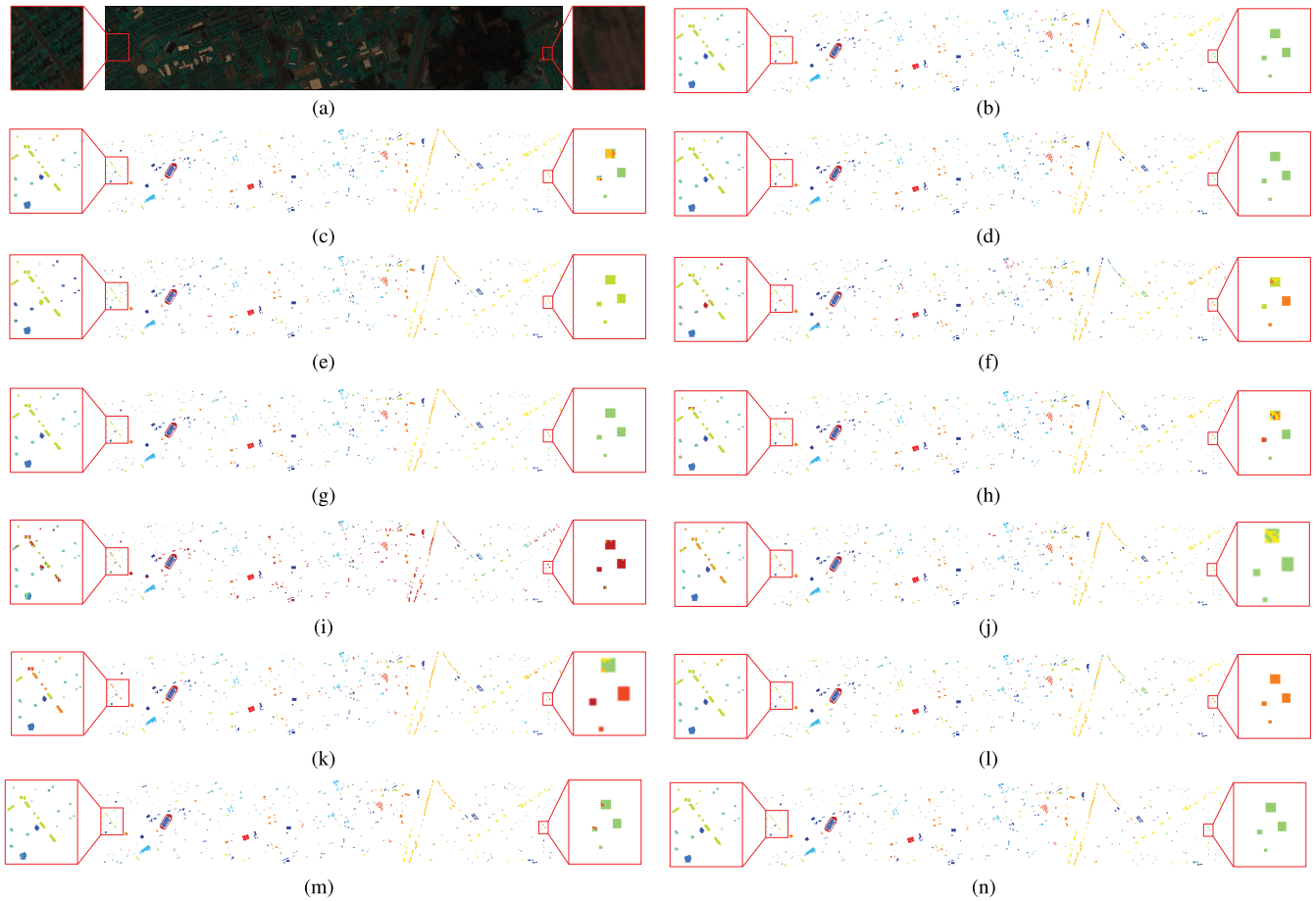

Fig. 7. Classification maps obtained by different methods on the *Houston University* dataset. (a) False-color image. (b) Ground-truth map. (c) SMR. (d) MSSGU. (e) SGL. (f) SSCL. (g) A^2S^2K. (h) ASSMN. (i) ADGAN. (j) MFL. (k) JSDF. (l) PKCRC. (m) IFRF. (n) ConGCN (proposed). In (a)–(n), zoomed-in views of the regions are denoted by red boxes.

more dominant in improving the model performance when compared with the spectral attention mechanism employed by A^2S^2K. Consequently, our proposed ConGCN achieves stable and encouraging performance.

The classification results produced by different methods are visualized in Fig. 7, where the zoomed-in regions are denoted by red boxes. As can be seen, there are noticeable errors in the classification maps of other methods. By contrast, our proposed ConGCN [Fig. 7(n)] achieves more precise classification results than other methods, which confirms the advantage of our method.

C. Impact of the Number of Labeled Examples

In this section, OAs of the proposed ConGCN and baseline methods under different numbers of labeled examples are shown in Fig. 8. We vary the number of labeled examples per class from 5 to 30 with an interval of 5 and report the OAs gained by all the methods on six datasets (i.e., *Indian Pines*, *University of Pavia*, *Salinas*, *Houston University*, *WHU-Hi-HanChuan*, and *WHU-Hi-HongHu* datasets). As observed in Fig. 8, the classification performances of all other methods except ADGAN can be generally improved by increasing the number of labeled examples. Due to the instability of GAN during training [62], the OAs of ADGAN could drop significantly on the *Indian Pines* [Fig. 8(a)], *University of Pavia* [Fig. 8(b)], *Salinas* [Fig. 8(c)], and *Houston University* [Fig. 8(d)] datasets even when the number of labeled examples increases. Because of the exploration of the supervision signals from both spectral and spatial aspects of HSI, our ConGCN still achieves relatively high OAs even if the labeled examples are quite limited (i.e., five or ten labeled examples per class), which suggests the good stability of ConGCN in HSI classification. The corresponding AAs and kappa coefficients under different numbers of labeled examples are presented in Section II-E of the Appendix. In addition, to verify the effectiveness of the proposed ConGCN when given limited labeled examples, we only use one, two, three, and four labeled examples per class for training, and the results are reported in Section II-F of the Appendix.

D. Ablation Study

As mentioned in Section I, the proposed ConGCN contains two parts that are critical for enriching the supervision signals from the spectral–spatial information of HSI, i.e., the semisupervised contrastive loss function and the graph generative loss function. We use the four datasets (i.e., *Indian Pines*, *University of Pavia*, *Salinas*, and *Houston University* datasets) to shed light on the contributions of these two components, where the number of labeled pixels per class is kept identical to the above experiments in Section VII-B. Every time we report the OA, AA, and kappa coefficient obtained by ConGCN without one of the aforementioned loss functions. For simplicity, "w/o CLoss" and "w/o GLoss" indicate the reduced models by removing the contrastive loss function $\mathcal{L}_{\text{ssc}}$ and the

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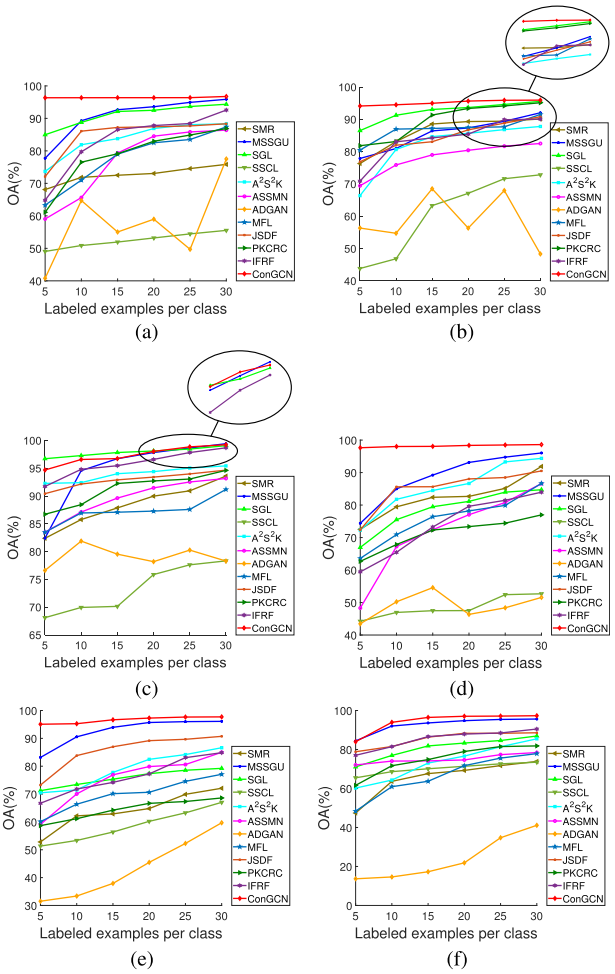

Fig. 8. OAs of various methods under different numbers of labeled examples per class. (a) *Indian Pines* dataset. (b) *University of Pavia* dataset. (c) *Salinas* dataset. (d) *Houston University* dataset. (e) *WHU-Hi-HanChuan* dataset. (f) *WHU-Hi-HongHu* dataset.

graph generative loss $\mathcal{L}_{g^2}$, respectively. In addition, we also investigate the effectiveness of the spatial-level and spectral-level graph augmentation. To be concrete, we utilize "w/o SpaAug" and "w/o SpeAug" to indicate the reduced models that remove the spatial-level graph augmentation and the spectral-level graph augmentation, respectively. Furthermore, we utilize "w/o CGSS" to indicate the reduced model that removes the semisupervised contrastive loss, graph generative loss, spatial-level graph augmentation, and spectral-level graph augmentation, simultaneously. Concretely, Tables VI–IX exhibit the comparative results, respectively. It is apparent that the OA will decline when any one of the aforementioned components is removed. Meanwhile, the reduced model, i.e., "w/o CGSS," obtains the lowest OA on each dataset. It reveals that each component makes an essential contribution to boosting the classification performance. We can also observe that the reduced model "w/o GLoss" consistently achieves higher OA, AA, and kappa coefficient than "w/o CLoss" on each dataset, which validates that the semisupervised contrastive loss makes a greater contribution than the generative loss to performance improvement.

VIII. Conclusion

In this article, we propose a ConGCN algorithm for HSI classification. To improve feature representation ability, we explore the supervision signals based on the spectral and spatial information of HSI. Specifically, we devise a semisupervised contrastive loss to exploit the supervision contained in the spectral signatures of image regions. Meanwhile, we develop a graph generative loss to explore supplementary supervision signals from the spatial relations among image regions. Last but not least, we devise an adaptive graph augmentation technique via incorporating the spectral–spatial priors to boost the performance of contrastive learning. As a consequence, the expressive power of the generated representation can be enhanced, which leads to the improved classification results. Experimental results on six real-world HSI datasets validate the effectiveness of our proposed ConGCN.

TABLE VI

OAs, AAs (%), and Kappa Coefficients Achieved by Different Settings on the *Indian Pines* Dataset. The Best and Second Best Records in Each Row Are **Bolded** and Underlined, Respectively

metrics	w/o CLoss	w/o GLoss	w/o SpaAug	w/o SpeAug	w/o CGSS	ConGCN
OA	96.44±0.55	96.46±0.68	96.58±0.82	96.56±0.68	95.78±1.33	**96.74±0.50**
AA	97.18±0.33	**97.28±0.37**	97.25±0.43	97.25±0.36	96.75±0.92	**97.28±0.29**
kappa	95.92±0.63	95.95±0.78	96.09±0.93	96.06±0.77	95.17±1.51	**96.27±0.57**

TABLE VII

OAs, AAs (%), and Kappa Coefficients Achieved by Different Settings on the *University of Pavia* Dataset. The Best and Second Best Records in Each Row Are **Bolded** and Underlined, Respectively

metrics	w/o CLoss	w/o GLoss	w/o SpaAug	w/o SpeAug	w/o CGSS	ConGCN
OA	95.05±1.13	95.32±1.54	95.37±1.62	95.79±1.64	94.62±1.36	**95.97±0.90**
AA	94.95±0.88	95.01±0.71	**95.70±0.39**	95.56±0.65	95.37±0.45	95.14±0.57
kappa	93.51±1.47	93.86±1.99	93.94±2.06	94.48±2.09	92.97±1.74	**94.69±1.17**

TABLE VIII

OAs, AAs (%), and Kappa Coefficients Achieved by Different Settings on the *Salinas* Dataset. The Best and Second Best Records in Each Row Are **Bolded** and Underlined, Respectively

metrics	w/o CLoss	w/o GLoss	w/o SpaAug	w/o SpeAug	w/o CGSS	ConGCN
OA	97.80±4.06	98.55±1.35	99.09±0.31	99.02±0.78	97.53±4.81	**99.25±0.29**
AA	98.50±2.06	98.90±0.74	99.20±0.13	99.07±0.42	98.59±1.43	**99.22±0.19**
kappa	97.56±4.48	98.39±1.50	98.99±0.35	98.91±0.86	97.28±5.29	**99.17±0.33**

TABLE IX

OAs, AAs (%), and Kappa Coefficients Achieved by Different Settings on the *Houston University* Dataset. The Best and Second Best Records in Each Row Are **Bolded** and Underlined, Respectively

metrics	w/o CLoss	w/o GLoss	w/o SpaAug	w/o SpeAug	w/o CGSS	ConGCN
OA	97.99±2.06	98.32±0.71	98.53±0.34	98.48±0.41	96.90±1.93	**98.61±0.37**
AA	97.93±2.25	98.33±0.69	98.53±0.29	98.49±0.38	96.83±2.04	**98.57±0.41**
kappa	97.82±2.22	98.18±0.77	98.41±0.37	98.36±0.44	96.64±2.09	**98.49±0.40**

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